

Probability density function

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Probability density function for discrete variable

Def

- In probability theory, a probability density function (pdf), or density of a continuous random variable is a function that describes the relative likelihood for this random variable to occur at a given point.
- The probability for the random variable to fall within a particular region is given by the integral of this variable's density over the region.
- The probability density function is non negative everywhere, and its integral over the entire space is equal to one. That is for absolutely continuous univariate distribution.

Some probability distribution of random variables (r.v)

- **Random variable** is the measurements of a random variable vary in a seemingly random and unpredictable manner.
- A random variable assumes a unique numerical value for each of the outcomes in the sample space of the probability experiment.
- If we toss a coin twice, the number of Heads obtained could be 0, 1 or 2.
- **Random Variable** is a variable whose values are determined by chance.

Example: Two Fair Coin Tosses

Let X = number of heads in two independent tosses of a fair coin. Possible outcomes: TT (0 heads), TH (1 head), HT (1 head), HH (2 heads)

$$P(\text{No heads}) = P(TT) = \left(\frac{1}{2}\right) \cdot \left(\frac{1}{2}\right) = \frac{1}{4} = 0.25$$

$$P(1 \text{ head}) = P(TH) + P(HT) = \left(\frac{1}{2} \cdot \frac{1}{2}\right) + \left(\frac{1}{2} \cdot \frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$P(2 \text{ heads}) = P(HH) = 0.5 \cdot 0.5 = 0.25$$

Probability Distribution Table

The probability distribution of X is often written as:

X	0	1	2
$P(X = x)$	0.25	0.5	0.25

Take away

- This is a valid discrete probability distribution: all probabilities are between 0 and 1, and sum to 1.
- $X \sim \text{Binomial}(n = 2, p = 0.5)$

Key Points

- $P(X = 0) = 0.25$
- $P(X = 1) = 0.50$
- $P(X = 2) = 0.25$
- Total probability = 1 $\rightarrow$ Valid **probability density function** (pdf) or **probability mass function** (pmf)

Definition of a Discrete Random Variable

Let X be a random variable that takes on values $x_1, x_2, \dots, x_n$ with corresponding probabilities:

$$P(X = x_1) = P_1$$

$$P(X = x_2) = P_2$$

$$\vdots$$

$$P(X = x_n) = P_n$$

Then, X is a **discrete random variable** if:

$$\sum_{i=1}^n P_i = 1 \quad \text{or} \quad \sum_{\text{all } x} P(X = x) = 1.$$

Notation convention:

- Random variables: denoted by capital letters: $X, Y, Z, \dots$
- Specific values they take: denoted by lowercase letters: $x, y, z, \dots$

Example: Two Fair Coin Flips

Let X = number of heads.

Sample space: TT, TH, HT, HH $\rightarrow$ each prob = $\frac{1}{4}$

$$P(X = 0) = P(TT) = 0.25$$

$$P(X = 1) = P(TH) + P(HT) = 0.5$$

$$P(X = 2) = P(HH) = 0.25$$

X	0	1	2
$P(X = x)$	0.25	0.5	0.25

Example: Probability Distribution of X

The following table gives the probability mass function (pmf) of a discrete random variable X :

i	1	2	3	4
x_i	-3	-2	1	4
$P(X = x_i)$	0.1	0.2	0.3	0.4

This means:

- $P(X = -3) = 0.1$
- $P(X = -2) = 0.2$
- $P(X = 1) = 0.3$
- $P(X = 4) = 0.4$

Note: The probabilities sum to 1 $\rightarrow$

$\sum P(X = x_i) = 0.1 + 0.2 + 0.3 + 0.4 = 1.0$, so this is a valid pmf.

Characteristics of a random variable.

Definition of Expected Value

Let X be a discrete random variable with probability mass function (pmf) $f(x) = P(X = x)$.

The **expectation** of X , written $E(X)$, is given by:

$$E(X) = \sum x \cdot P(X = x) \quad \text{or} \quad E(X) = \sum_{i=1}^n x_i \cdot p_i \quad \text{or} \quad E(X) = \sum_{i=1}^n x_i \cdot f(x_i)$$

This represents the **weighted average** of all possible values of X , where weights are their respective probabilities.

Interpretation: Gains and Losses

The expected value can be interpreted as:

- The **average gain or loss** per play in a game of chance, if played many times.
- A long-run average outcome, weighted by probabilities.

For example:

- If $E(X) > 0$: the game is **favorable** (expected gain).
- If $E(X) < 0$: the game is **unfavorable** (expected loss).
- If $E(X) = 0$: the game is **balanced** (fair game).

Take away

Expected Value = Weighted Average of Outcomes

$$E(X) = \sum x_i \cdot P(X = x_i)$$

Game Interpretation

- $E(X) > 0 \rightarrow$ Favorable
- $E(X) < 0 \rightarrow$ Unfavorable
- $E(X) = 0 \rightarrow$ Balanced (Fair)

This distribution is symmetric — expected value
 $E(X) = 1$

Properties of Expected Value

1. Scaling and Shifting: Let X be a random variable and k a real number.

$$E(kX) = k \cdot E(X)$$

$$E(X + k) = E(X) + k$$

These show that expectation is linear with respect to scaling and shifting.

2. Additivity: Let X and Y be two random variables (discrete or continuous).

$$E(X + Y) = E(X) + E(Y)$$

This holds even if X and Y are dependent — a key property of expectation!

Summary: Linearity of Expectation

The expected value operator $E(\cdot)$ is **linear**, meaning:

$$E(aX + bY + c) = aE(X) + bE(Y) + c$$

for any constants a, b, c and random variables X, Y .

Important Note

Linearity holds regardless of whether X and Y are independent or not.

Example: Find $E(X)$

A discrete random variable X has the following probability distribution:

X	-2	-1	0	1	2
$P(X = x)$	0.3	0.1	0.15	0.4	0.05

Solution: Compute $E(X)$

The expected value is given by:

$$E(X) = \sum_{\text{all } x} x \cdot P(X = x)$$

Substitute the values:

$$\begin{aligned} E(X) &= (-2)(0.3) + (-1)(0.1) + (0)(0.15) + (1)(0.4) + (2)(0.05) \\ &= -0.6 - 0.1 + 0 + 0.4 + 0.1 \\ &= -0.2 \end{aligned}$$

$$E(X) = -0.2$$

Interpretation

- The expected value $E(X) = -0.2$ means that, on average, the outcome is slightly negative.
- If this represents a game or payoff, you would expect to lose 0.2 units per trial in the long run.

Key Takeaway

Expected value is a weighted average — each outcome is multiplied by its probability before summing.

The Expectation of Any Function of X , $\mathbb{E}[g(X)]$

Let X be a r.v. and pdf $f(x) = P(X = x)$. Let also $g(x)$ be a function of the r.v. X . Then, the expected value of $g(x)$, written $\mathbb{E}[g(X)]$, is given by:

$$\mathbb{E}[g(X)] = \sum_{\text{all } x} g(x) P(X = x)$$

Example: Expectation of Discrete Random Variable

The r.v. X has pdf $P(X = x)$ for $x = 1, 2, 3$.

X	1	2	3
$P(X = x)$	0.1	0.6	0.3

Compute:

- a) $E(3)$
- b) $E(X)$
- c) $E(5X)$
- d) $E(5X + 3)$

Solution

- a) $E(3) = \sum_{\text{all } x} 3 \cdot P(X = x) = 3 \sum_{\text{all } x} P(X = x) = 3 \cdot (0.1 + 0.6 + 0.3) = 3 \cdot 1 = 3$
- b) $E(X) = \sum_{\text{all } x} x \cdot P(X = x) = (1 \cdot 0.1) + (2 \cdot 0.6) + (3 \cdot 0.3) = 0.1 + 1.2 + 0.9 = 2.2$
- c) $E(5X) = \sum_{\text{all } x} 5x \cdot P(X = x) = (5 \cdot 1 \cdot 0.1) + (5 \cdot 2 \cdot 0.6) + (5 \cdot 3 \cdot 0.3) = 0.5 + 6.0 + 4.5 = 11$
- d) $E(5X + 3) = \sum_{\text{all } x} (5x + 3) \cdot P(X = x) = (8 \cdot 0.1) + (13 \cdot 0.6) + (18 \cdot 0.3) = 0.8 + 7.8 + 5.4 = 14$

Variance ($\text{Var } X$) and Standard Deviation σ_X

Let X be a r.v. having the pdf $f(x) = P(X = x)$, $x \in \mathbb{R}$, and let $E(X) = \mu$, where μ is constant.

The variance of X , written $\text{Var}(X)$, is given by:

$$\text{Var}(X) = E[(X - \mu)^2]$$

Alternatively,

Variance: Derivation

$$\begin{aligned}\text{Var}(X) &= E[(X - \mu)^2] \\&= E[X^2 - 2\mu X + \mu^2] \\&= E(X^2) - 2\mu \cdot E(X) + E(\mu^2) \\&= E(X^2) - 2\mu \cdot \mu + \mu^2 \quad (\text{since } \mu \text{ is constant}) \\&= E(X^2) - 2\mu^2 + \mu^2 \\&= E(X^2) - \mu^2 \\&= E(X^2) - [E(X)]^2\end{aligned}$$

For a discrete random variable taking values $x_1, x_2, \dots, x_n$ with probabilities $P_i = P(X = x_i)$:

$$\text{Var}(X) = \sum_{i=1}^n x_i^2 \cdot P_i - [E(X)]^2$$

The standard deviation is:

$$\sigma_X = \sqrt{\text{Var}(X)}$$

Properties of Variance and Standard Deviation

- $\text{Var}(k) = 0$ (variance of a constant is zero)
- $\text{Var}(kX) = k^2 \text{Var}(X)$
- $\text{Var}(X + k) = \text{Var}(X)$ (adding a constant doesn't change variance)
- $\text{Var}(kX + b) = k^2 \text{Var}(X)$
- $\sigma(kX) = \sigma_{kX} = |k| \cdot \sigma_X$

Example: Computing $E(X)$, $\text{Var}(X)$, and σ_X

An r.v. X has the following distribution:

X	1	2	3	4	5
$P(X = x)$	0.1	0.3	0.2	0.3	0.1

Find $E(X)$, $\text{Var}(X)$, and σ_X .

Solution

a) $E(X) = \sum_{\text{all } x} x \cdot P(X = x) =$
 $(1 \cdot 0.1) + (2 \cdot 0.3) + (3 \cdot 0.2) + (4 \cdot 0.3) + (5 \cdot 0.1) = 3$

b) First compute $E(X^2)$:

$$E(X^2) = \sum_{\text{all } x} x^2 \cdot P(X = x) = (1^2 \cdot 0.1) + (2^2 \cdot 0.3) + (3^2 \cdot 0.2) + (4^2 \cdot 0.3) + (5^2 \cdot 0.1) = 10.4$$

Then,

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 10.4 - 3^2 = 1.4$$

$$\sigma_X = \sqrt{\text{Var}(X)} = \sqrt{1.4} \approx 1.1832$$

Properties

For a and b constants and X and Y random variables.

- ① $E[b] = b$;
- ② $E[aX] = aE[X]$;
- ③ $E[aX + b] = aE[X] + b$;
- ④ $E[X + Y] = E[X] + E[Y]$;
- ⑤ $E[XY] = E[X]E[Y]$; for X and Y independent.

Variance

The variance of a random variable X is given by

$$\sigma^2 = E[(X - \mu)^2] = E(X^2) - E^2(X)$$

For the continuous case the proof is step by step the same, with summations replaced by integrations.

Exercise 1

Let the random variable X represent the number of defective parts for a machine when 3 parts are sampled from a production line and tested. The following is the probability distribution of X . Calculate the variance σ^2 .

x	0	1	2	3
$f(x)$	0.51	0.38	0.10	0.01